

Recommendations — urothelial-mets-her2-discordant-kndl

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** bladder (urothelial); separate second primary: prostate, treated 2020, on surveillance with no evidence of disease
- **Histology:** high-grade muscle-invasive urothelial carcinoma, NOS, poorly differentiated (focal areas suggestive of plasmacytoid morphology at diagnosis; pleomorphic giant cells in 2025 recurrence; trophoblastic differentiation excluded by HCG IHC; no definitive micropapillary features on expert re-read)
- **Stage:** IVB (cT3 cN0 pM1b at diagnosis 2023-11 with liver metastases; CR on 1L therapy 2024-08; current disease is oligometastatic nodal — biopsy-proven left para-aortic node plus probable perivesical node, treated with SBRT 2026-06)
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** male
- **Biomarkers:** HER2 positive (IHC 3+) — three concordant validated results: primary bladder tumor 2023-11, liver metastasis 2023-11, bladder recurrence 2025-03; DISCORDANT weak HER2 signal on research-grade multiplex immunofluorescence (MXIF) of recent lymph-node tissue 2026-08; repeat validated IHC performed 2026-08, tier not yet in available record (ihc_pending); PD-L1 (CPS) CPS 5 / tumor cells 5% (primary bladder 2023-11); CPS 0 (liver metastasis 2023-11) — discordant primary vs metastasis; Nectin-4 highly positive: 95% membranous tumor expression (primary, 2023-11); 100% membranous in in-situ component (2025-03 recurrence). Current expression after ~25 months of enfortumab vedotin is unknown; Nectin-4-directed imaging performed 2026-06 (result not in available record); TROP2 low positive (weak) on research-grade MXIF 2026-08 — recorded as low expression, NOT negative; validated TROP2 IHC performed 2026-08, tier not yet in available record (ihc_pending); Mesothelin strong expression on research-grade MXIF (2026-08) (ihc_pending); MUC1 strong expression on research-grade MXIF (2026-08) (ihc_pending); FAP (stromal) FAP-positive stroma on research-grade MXIF (2026-08); prior FAPi PET (2025-09) showed intense FAP avidity in prostate/seminal vesicle only — confounded by biopsy-proven BCG granulomatous inflammation; no FAP-avid urothelial carcinoma on that scan; repeat FAP imaging 2026-06 (result not in available record) (unknown); PRAME PRAME-directed imaging completed 2026-08; result not in available record (unknown); MAGE-A4 tumor expression negative (2024-06); patient was HLA-matched for a MAGE-A4 TCR trial but expression testing forecloses it; TMB 12.4 mut/Mb in both primary bladder tumor and liver metastasis (institutional median 3.9; bladder-cancer median 8.8); MSI microsatellite stable (MSS), both specimens; Somatic mutations (named in available record) ARID1A, TERT promoter, TP53 (incl. TP53 R175H seen in plasma ctDNA at 0.09% VAF, 2025-06) — recurrently detected in urine ctDNA (UroAmp) and concordant with tumor tissue; tissue NGS summary: 16 mutations + 12 copy-number alterations (primary), 15 mutations + 13 CNAs (liver met), no structural variants, exactly 1 alteration with an OncoKB treatment interpretation — the identity of that actionable alteration is NOT in the available record; HLA class I HLA-A*02:01 positive (high-resolution class I and II typing on file in local record; full typing withheld from this derived profile as quasi-identifying); Personalized neoantigens / vaccine immunogenicity 24 predicted neoepitopes selected from tumor WES (2023-11; class I 8-12mer and class II ~17mer peptides); EliSpot at 4 months post-vaccination: CD4+ T-cell responses to 2 of 14 evaluable peptide pools, no responses at day 1, several pools not evaluable; serial pre/post-vaccine PBMCs banked for ongoing EliSpot monitoring; ctDNA (tumor-informed MRD) Signatera: chronically positive 0.04-1.76 MTM/mL from 2024-05, rising to 3.29-3.45 MTM/mL 2026-06, then 0 MTM/mL on two draws 2026-06/2026-07 (per ops summary; the two zero-draws are also listed as pending in one table — minor internal conflict). NeXT

Personal: 813 PPM (2026-06), peak 848 PPM (2026-05), with a reported decrease after an initial transient post-therapy increase (2026-08 report)

- **Efficacy/toxicity weight:** 0.7
- **Toxicity vetoes:** severe peripheral neuropathy (INFERRED: established CIPN from enfortumab vedotin with repeated dose reductions and an active CIPN-management project — avoid or de-prioritize high-neuropathy agents, especially MMAE-payload ADCs and platinum/taxane backbones, where an alternative exists), severe GI toxicity / highly emetogenic regimens (INFERRED: severe nausea and bloating after trastuzumab deruxtecan dose 2 is an active, dose-limiting problem), interstitial lung disease / pneumonitis-high-risk agents without tight monitoring (INFERRED: 2026-03 outside CT read suggestive of early pulmonary fibrosis/ILD; directly interacts with T-DXd ILD risk), pancreatitis-risk agents (INFERRED: recovered asymptomatic pancreatitis 2025 with lipase to ~3x ULN on EV/pembro; amylase/lipase now monitored q3-4wk)
- **Modality constraints:** no geographic or logistical constraints evident (INFERRED: patient has traveled internationally for imaging, vaccination, and consults, and maintains care at a US-Northeast tertiary academic center), intensive/experimental modalities acceptable, including cell therapy requiring apheresis (INFERRED: apheresis already banked; adoptive cell therapy (TIL), NK-cell plus N-803 single-patient IND, and vaccine programs actively pursued)

53 ranked therapeutic options across 12 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Her2 IHC interventions

1. zanidatamab (HER2 biparatopic bispecific, payload-free) — off-label or NCT06695845 fallback if 3+ holds but T-DXd stops *(conditional on her2_ihc positive)*

Likelihood of effect: Moderate if 3+ holds: ORR 37.0% (95% CI 27.0-48.7) in phase 1 (pmid:36400106), biliary-weighted; untested after T-DXd.

Toxicity burden: Low (grade-3 TRAEs in 4/132, no treatment-related deaths; LVEF monitoring)

Counter-productive MoA:Low (ADC-selected HER2 downregulation would blunt a payload-free bispecific; untested after T-DXd.)

Overall:The payload-free way to stay on a confirmed HER2 3+ axis when T-DXd stops for tolerability — clean safety file, thin urothelial evidence, trial door closed by prior T-DXd.

Key references: PMID 36400106 · NCT06695845 · NCT02892123

2. pertuzumab + trastuzumab (payload-free HER2 doublet) — TAPUR HER2 arm NCT02693535 or off-label, pending the ERBB2 report *(conditional on her2_ihc positive)*

Likelihood of effect: Moderate at amplified 3+ (ORR 41.0%, MyPathway pmid:37793085); collapses to 8.3% at 1+; untested after a HER2 ADC.

Toxicity burden: Low (diarrhea 36.1% mostly low-grade; no serious term above 1.7%; LVEF monitoring)

Counter-productive MoA:Low (ADC-selected HER2 loss would gut a naked-antibody doublet; the tier gradient is

steep.)

Overall:The one HER2 route that touches none of the four vetoes, with a protocolized TAPUR path — its floor set by the pending tier and the unread ERBB2 copy number.

Key references: PMID 37793085 · NCT02693535

3. FT825 / ONO-8250 (off-the-shelf iPSC-derived HER2 CAR-T) — NCT06241456

(conditional on her2_ihc positive)

Likelihood of effect: Unknown — phase 1 with no urothelial signal; benefit is expected to track the pending HER2 tier (pmid:37870536).

Toxicity burden: High (lymphodepletion cytopenias and CRS; no mature safety table)

Counter-productive MoA:Low (Lymphodepletion could blunt the concurrently primed vaccine T-cell axis.)

Overall:The HER2 cell-therapy route that survives the file's constraints — no apheresis, no deruxtecan payload — and nothing measured in urothelial cancer yet.

Key references: NCT06241456 · PMID 37870536 · PMID 37793085

4. trastuzumab + paclitaxel (HER2 doublet with taxane backbone) — off-label

Likelihood of effect: Moderate response plausible at 3+ but irrelevant: the taxane backbone lands on the stated neuropathy veto (pmid:37385153).

Toxicity burden: Moderate (any-grade neuropathy 88.9% from the taxane backbone — the vetoed toxicity class)

Counter-productive MoA:Low (None mechanism-level; the disqualifier is the taxane-neuropathy collision.)

Overall:An accessible HER2 doublet disqualified by its backbone: 88.9% any-grade neuropathy against established CIPN.

Key references: PMID 37385153

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

disitamab vedotin (RC48, HER2 ADC, MMAE payload) — NCT04879329 and US cohorts — _Not enrollable_

Likelihood of effect: Moderate on the published record (ORR 50.5%, pmid:37988648) but unreachable: every US cohort excludes this history.

Toxicity burden: High (sensory neuropathy 68.2% any-grade, 18.7% grade ≥ 3 on an MMAE payload)

Counter-productive MoA:Low (None assessed; exclusion criteria and the MMAE-CIPN collision are the block.)

Overall:The strongest HER2 number in the disease, foreclosed twice over: MMAE against established CIPN, and US exclusions for prior HER2 and MMAE therapy.

Key references: PMID 37988648 · NCT04879329

CAdVEC oncolytic adenovirus + HER2 CAR virus-specific T cells (intratumoral) — NCT03740256 — _Thin evidence_

Likelihood of effect: Unknown: no published efficacy data; injectable-lesion and single-site constraints on top.

Toxicity burden: Moderate (oncolytic + CAR-T combination; CRS and local injection risks, unquantified)

Counter-productive MoA:Low (None assessed beyond class uncertainty.)

Overall:The one HER2 row that tolerates a 2+ tier — behind an injectable-lesion gate and a single Texas site.

Key references: NCT03740256

Evidence in detail

zanidatamab (HER2 biparatopic bispecific, payload-free) — off-label or NCT06695845 fallback if 3+ holds but T-DXd stops

****Meric-Bernstam et al. *Lancet Oncol* 2022****

- **Disease context:**Locally advanced or metastatic HER2-expressing or HER2-amplified solid tumors of any type after all approved therapies (2L+) — Part 2 enrolled biliary tract (n=22), colorectal (n=28) and other HER2-expressing/amplified cancers excluding breast and gastroesophageal (n=36).
- **n:** 132
- **Toxicities:**diarrhoea (grade any): 43.0% (n=86); **infusion-related reaction** (grade any): 34.0% (n=86); **any treatment-related AE** (grade 3): 3.0% (n=132); **treatment-related death** (grade 5): 0.0% (n=132)
- **Efficacy:**ORR 37.0% (95% CI 27.0-48.7); **AE_rate** 3.0%

pertuzumab + trastuzumab (payload-free HER2 doublet) — TAPUR HER2 arm NCT02693535 or off-label, pending the ERBB2 report

****Sweeney et al. *J Clin Oncol* 2024****

- **Disease context:**HER2-amplified and/or overexpressed and/or mutated solid tumors, tissue-agnostic cohort (2L+) — 263 patients with HER2 amplification and/or overexpression; 83 with HER2 mutation alone. Activity was stratified by KRAS status and by HER2 IHC tier.
- **n:** 346
- **Toxicities:**diarrhoea (grade any): 36.1% (n=357); **fatigue** (grade any): 29.4% (n=357); **nausea** (grade any): 22.4% (n=357); **anaemia** (grade any): 17.6% (n=357); **infusion-related reaction** (grade any): 14.3% (n=357); **any serious adverse event** (grade any): 26.6% (n=357); **small intestinal obstruction** (grade any): 1.7% (n=357)
- **Efficacy:**ORR 25.9% (95% CI 20.7-31.6); **ORR** 41.0%; **ORR** 21.9%; **ORR** 8.3%; **ORR** 0.0%; **ORR** 28.1%; **ORR**

6.0% (95% CI 2.0-13.5)

FT825 / ONO-8250 (off-the-shelf iPSC-derived HER2 CAR-T) — NCT06241456

[Meric-Bernstam et al. J Clin Oncol 2024](#)

- **Disease context:**HER2-expressing locally advanced or metastatic solid tumors (seven cohorts; bladder cohort n=41), excluding breast, gastric, colorectal and NSCLC (2L+) — HER2 IHC 3+ or 2+ by local or central testing, progression after ≥1 systemic therapy or no satisfactory alternative; ECOG 0-1. Bladder cohort was metastatic or advanced urothelial carcinoma of renal pelvis, ureter, bladder or urethra. Prior HER2-directed therapy was permitted.
- n: 267
- **Toxicities:**any drug-related AE (grade any): 84.6% (n=267); any drug-related AE (grade ≥3): 40.8% (n=267); nausea (grade any): 55.1% (n=267); vomiting (grade any): 24.7% (n=267); diarrhea (grade any): 25.8% (n=267); anemia (grade any): 27.7% (n=267); fatigue (grade any): 24.7% (n=267); ILD/pneumonitis (adjudicated drug-related) (grade any): 10.5% (n=267); ILD/pneumonitis (adjudicated drug-related) (grade ≥3): 1.5% (n=267); serious drug-related AE (grade any): 13.5% (n=267)
- **Efficacy:**ORR 39.0% (95% CI 24.2-55.5); ORR 41.5% (95% CI 26.3-57.9); ORR 56.3% (95% CI 29.9-80.2); ORR 61.3% (95% CI 49.4-72.4); ORR 37.1% (95% CI 31.3-43.2); ORR 36.8% (95% CI 21.8-54.0); DoR 11.3 months (95% CI 9.6-17.8); PFS 6.9 months (95% CI 5.6-8.0); OS 13.4 months (95% CI 11.9-15.5); AE_rate 40.8%; discontinuation_rate 8.6%

[Sweeney et al. J Clin Oncol 2024](#)

- **Disease context:**HER2-amplified and/or overexpressed and/or mutated solid tumors, tissue-agnostic cohort (2L+) — 263 patients with HER2 amplification and/or overexpression; 83 with HER2 mutation alone. Activity was stratified by KRAS status and by HER2 IHC tier.
- n: 346
- **Toxicities:**diarrhoea (grade any): 36.1% (n=357); fatigue (grade any): 29.4% (n=357); nausea (grade any): 22.4% (n=357); anaemia (grade any): 17.6% (n=357); infusion-related reaction (grade any): 14.3% (n=357); any serious adverse event (grade any): 26.6% (n=357); small intestinal obstruction (grade any): 1.7% (n=357)
- **Efficacy:**ORR 25.9% (95% CI 20.7-31.6); ORR 41.0%; ORR 21.9%; ORR 8.3%; ORR 0.0%; ORR 28.1%; ORR 6.0% (95% CI 2.0-13.5)

trastuzumab + paclitaxel (HER2 doublet with taxane backbone) — off-label

[Kim et al. ESMO Open 2023](#)

- **Disease context:**HER2-positive recurrent or metastatic urothelial carcinoma (2L+) — HER2 positivity by IHC per 2013 ASCO/CAP breast criteria; 26 of 27 enrolled patients were evaluable.
- n: 27
- **Toxicities:**peripheral neuropathy (grade any): 88.9% (n=27); neutropenia (grade ≥3): 25.9% (n=27); thrombocytopenia (grade ≥3): 7.4% (n=27); anemia (grade ≥3): 7.4% (n=27)
- **Efficacy:**ORR 48.1%; DoR 6.9 months (95% CI 4.4-9.3); PFS 8.4 months (95% CI 6.2-8.8); OS 13.5 months

disitamab vedotin (RC48, HER2 ADC, MMAE payload) — NCT04879329 and US cohorts

[Sheng et al. J Clin Oncol 2024](#)

- **Disease context:**HER2-positive (IHC 3+ or 2+) locally advanced or metastatic urothelial carcinoma (2L+) — Pooled RC48-C005 and RC48-C009; all patients had progressed on ≥ 1 prior systemic chemotherapy line. Subgroups included liver metastases and prior anti-PD-1/L1 exposure.
- **n:** 107
- **Toxicities:**peripheral sensory neuropathy (grade any): 68.2% (n=107); peripheral sensory neuropathy (grade ≥ 3): 18.7% (n=107); leukopenia (grade any): 50.5% (n=107); neutropenia (grade any): 42.1% (n=107); neutropenia (grade ≥ 3): 12.1% (n=107); **AST increased** (grade any): 42.1% (n=107); **any treatment-related AE** (grade ≥ 3): 54.2% (n=107)
- **Efficacy:**ORR 50.5% (95% CI 40.6-60.3); **DoR** 7.3 months (95% CI 5.7-10.8); **PFS** 5.9 months (95% CI 4.3-7.2); **OS** 14.2 months (95% CI 9.7-18.8); **TRAE_rate** 54.2%

Nectin4 interventions

1. [225Ac]Ac-AKY-1189 (Nectin-4 actinium-225 radioligand) with [64Cu]Cu-AKY-1189 imaging — NCT07020117

Likelihood of effect: Unknown — first-in-human with zero efficacy data; a mechanism bet that alpha crossfire survives partial Nectin-4 loss after 25 months of EV (pmid:41173324).

Toxicity burden: Moderate (no human safety table; alpha-radioligand class marrow suppression and renal exposure, with eGFR unmeasured)

Counter-productive MoA:Low (Subtherapeutic dose-escalation assignment could spend the post-T-DXd window without benefit.)

Overall:The one Nectin-4 route insulated from both MMAE neuropathy and antigen heterogeneity, with an imaging step that answers the expression question tissue-free — and no efficacy data at all.

Key references: NCT07020117 · PMID 41173324 · PMID 40931013

2. LY4052031 (Nectin-4 ADC, camptothecin topoisomerase-I payload) — NEXUS-01 cohort B2, NCT06465069

Likelihood of effect: Moderate but unstable: 47% post-EV ORR is unpublished dose-escalation data (NCT06465069); target validation is RCT-grade, the agent estimate is not.

Toxicity burden: Moderate (camptothecin-class diarrhea, nausea, neutropenia; no published safety table yet)

Counter-productive MoA:Moderate (Sequential topoisomerase-I payloads after T-DXd risk cross-resistance; current Nectin-4 expression unverified.)

Overall:The trial written for his exact post-EV history with a payload that clears the neuropathy veto — carried by RCT-grade target validation, not by published data on this drug.

Key references: NCT06465069 · PMID 42167230 · PMID 38446675

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

AVZO-103 (Nectin-4 x TROP2 bispecific ADC, exatecan payload) — NCT07193511 — _Not enrollable_

Likelihood of effect: Unassessable until the chest CT discrepancy is adjudicated; preclinical bispecific design suits heterogeneous antigen (pmid:41101697).

Toxicity burden: Moderate (exatecan-payload class GI and marrow toxicity; no clinical safety table)

Counter-productive MoA:Low (None assessed; enrollability, not mechanism, is the block.)

Overall:The dual-target ADC two personas wanted to rank — blocked solely by an ILD exclusion against an unresolved CT read.

Key references: NCT07193511 · PMID 41101697

CRB-701 / SYS6002 (site-specific Nectin-4 ADC, MMAE payload) — NCT06265727 — _Consolidated_

Likelihood of effect: See rank 5: the approach is ranked once; this product keeps the vetoed payload class.

Toxicity burden: High (MMAE-class neuropathy risk against established CIPN)

Counter-productive MoA:Low (Same antigen-dependency as the ranked approach, plus the MMAE liability.)

Overall:A tidier linker on the payload the file vetoes; the ranked Nectin-4 ADC changed the payload instead.

Key references: NCT06265727

E303 / SBE303 (Nectin-4 ADC) — NCT07524348 — _Consolidated_

Likelihood of effect: See rank 5: the approach is ranked once; no data distinguish this product.

Toxicity burden: Moderate (ADC class profile; payload details thin in the record)

Counter-productive MoA:Low (Same antigen-dependency as the ranked approach.)

Overall:The backup sponsor for the rank-5 approach if NEXUS-01 screening fails.

Key references: NCT07524348

CT-202 (Nectin-4-directed bispecific antibody, payload-free) — NCT07545122 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; same current-expression question as every Nectin-4 row.

Toxicity burden: Low (no payload; bispecific-class infusion and immune events, unquantified)

Counter-productive MoA:Low (Payload-free killing depends more, not less, on retained antigen density.)

Overall:The gentlest Nectin-4 concept on paper, with nothing measured behind it yet.

Key references: NCT07545122

EB-DT-NK-UC101 (cord-blood Nectin-4/HER2 dual-target CAR-NK) — NCT07492628 — _Not enrollable_

Likelihood of effect: Not applicable: not accessible from the US today.

Toxicity burden: High (lymphodepletion plus CAR-NK class toxicity, unquantified)

Counter-productive MoA:Low (None assessed; access is the block.)

Overall:The most permissive eligibility on the list, attached to a program he cannot reach.

Key references: NCT07492628

Nectin-4-directed CAR T cells (preclinical) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: High (CAR-T class, hypothetically)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 40931013

ETx-22 (Nectin-4 ADC, exatecan payload, preclinical) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (exatecan class, hypothetically)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 39440991

Nectin-4-targeted near-infrared photoimmunotherapy (preclinical) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (local phototoxicity, hypothetically)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Evidence in detail

LY4052031 (Nectin-4 ADC, camptothecin topoisomerase-I payload) — NEXUS-01 cohort B2, NCT06465069

Powles et al. *N Engl J Med* 2024

- **Disease context:** Previously untreated locally advanced or metastatic urothelial carcinoma (1L) — Cisplatin-eligible and cisplatin-ineligible patients both enrolled; co-primary endpoints PFS by BICR and OS.
- **n:** 886
- **Toxicities:** any treatment-related AE (grade ≥ 3): 55.9% (n=440)
- **Efficacy:** OS 31.5 months; HR_OS 0.47 (95% CI 0.38-0.58); PFS 12.5 months; HR_PFS 0.45 (95% CI 0.38-0.54); TRAE_rate 55.9%

Tp53 R175H interventions

1. CLSP-1025 (TP53 R175H x HLA-A*02:01 T-cell engager) — GUARDIAN-101, NCT06778863

Likelihood of effect: Unknown, and gated twice: no clinical data for the class, and the R175H call is one 0.09% VAF plasma read pending tissue confirmation.

Toxicity burden: Moderate (expected CRS with inpatient step-up dosing; no human safety table yet)

Counter-productive MoA: Moderate (Very low peptide-HLA density and unmeasured HLA-A*02 LOH could cap engager potency.)

Overall: A truncal-driver bet with both eligibility gates already on file — but the R175H call is a single 0.09% VAF plasma read and the class has no clinical results yet.

Key references: NCT06778863 · PMID 33649166 · PMID 41542775

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

NT-175 (autologous TCR-T targeting TP53 R175H) — NCT05877599 — _Consolidated_

Likelihood of effect: See rank 4: same axis and same 0.09% VAF hedge; adds lymphodepletion and a manufacturing wait.

Toxicity burden: High (fludarabine/cyclophosphamide lymphodepletion plus TCR-T class CRS)

Counter-productive MoA: Moderate (Same low peptide-HLA density obstacle as the ranked engager.)

Overall: The autologous fallback for the rank-4 axis — same gates, more toxicity, longer runway.

Key references: NCT05877599 · PMID 35749374 · PMID 31996390

Fgfr3 interventions

1. FGFR3 axis, conditional on the sequencing retrieval: vepugratinib (LOXO-435) FORAGER-1 NCT05614739 as the investigational arm (erdafitinib, if an alteration is found, is the approved default and belongs to the standard-of-care table)

Likelihood of effect: Conditional: ~15-20% chance a qualifying FGFR3 alteration exists; if found, both guideline bodies endorse the axis in exactly this line (NCT05614739).

Toxicity burden: Moderate (FGFR-class hyperphosphatemia, stomatitis, nail and eye toxicity; no agent-specific table in the dossier)

Counter-productive MoA:Low (None plausible before the gate; expected yield is the limiter.)

Overall:A records request away from being real: FGFR3 status is simply unread, and the FGFR axis is the only non-HER2 route both major guidelines endorse in this line.

Key references: NCT05614739 · NCT03390504

Trop2 interventions

1. ASP2998 (TROP2-directed STING agonist conjugate) — NCT07287995, monotherapy or pembrolizumab arm; EV combination arms declined in writing

Likelihood of effect: Low-to-unknown: no clinical data, and low-positive TROP2 is a weaker predictor even where entry is expression-agnostic (pmid:39086310).

Toxicity burden: Moderate (no human data; immune-agonist class inflammatory events with a pancreatitis-history watch)

Counter-productive MoA:Low (Weak TROP2 may limit conjugate delivery regardless of payload class.)

Overall:The TROP2 door that stays open after T-DXd — expression-agnostic entry, with the low-positive hedge stated and the MMAE combination arms declined in writing.

Key references: NCT07287995 · PMID 39086310

2. sacituzumab tirumotecan (sac-TMT) phase 3 randomized trial — NCT07419295 (TROP2 axis via randomization)

Likelihood of effect: Not applicable as ranked: ineligible (no prior platinum) and the randomization veto stands; class OS evidence is negative (pmid:39934055).

Toxicity burden: High (potential taxane-arm CIPN on randomization; class grade-5 TEAEs 7% in TROPiCS-04)

Counter-productive MoA:High (Taxane-arm assignment would inflict the vetoed neuropathy and could end all

ADC options after.)

Overall:Randomized entry is the guideline-clean route to TROP2, but this trial can assign a taxane he cannot take and requires platinum he never had.

Key references: NCT07419295 · PMID 39934055 · PMID 39086310

3. sacituzumab govitecan (TROP2 ADC) — trial-only after the 2024 urothelial withdrawal (NCT03547973 cohort 2 is the platinum-ineligible route)

Likelihood of effect: Low: the confirmatory phase 3 missed OS (HR 0.86, p=0.087) and the urothelial approval was withdrawn (pmid:39934055).

Toxicity burden: High (grade-5 TEAEs 7% in TROPiCS-04, mostly neutropenic infection; diarrhea, neutropenia)

Counter-productive MoA:Moderate (Response without survival — the surrogate trap; low TROP2 weakens the predictor further.)

Overall:The TROP2 class's own confirmatory test came back negative in this disease; a slot spent here competes with better-fitted doors.

Key references: PMID 39934055 · PMID 33929895 · NCT03547973

4. datopotamab deruxtecan (Dato-DXd, TROP2 ADC, deruxtecan payload) — urothelial substudy of NCT05489211

Likelihood of effect: Unknown in urothelial carcinoma; low-positive TROP2 and back-to-back deruxtecan payloads argue against prioritizing it.

Toxicity burden: Moderate (deruxtecan-class nausea, stomatitis and ILD risk — the toxicities already limiting T-DXd)

Counter-productive MoA:Low (Sequential deruxtecan payloads invite cross-resistance and cumulative ILD exposure.)

Overall:Reaches the TROP2 axis only by re-running the exact payload toxicities that are already dose-limiting on the current drug.

Key references: NCT05489211 · PMID 40862536

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

LCB84 (TROP2-directed ADC) — NCT05941507 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; low-positive TROP2 is a weak predictor for the class.

Toxicity burden: Moderate (ADC class GI and marrow toxicity; no safety table)

Counter-productive MoA:Low (Weak antigen may limit delivery, as across the TROP2 axis.)

Overall:A backup TROP2 door with permissive entry and no data.

Key references: NCT05941507

R7059-DXd (epitope-distinct TROP2 ADC, preclinical) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (DXd class, hypothetically)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Evidence in detail

ASP2998 (TROP2-directed STING agonist conjugate) — NCT07287995, monotherapy or pembrolizumab arm; EV combination arms declined in writing

Loriot et al. *Clin Cancer Res* 2024

- **Disease context:**Unresectable locally advanced or metastatic urothelial carcinoma (2L+) — Archival tumor from TROPY-U-01 cohorts 1-3; 146 of 192 treated patients had evaluable TROP2 data scored by H-score (0-300) and percentage membrane-positive tumor cells.
- **n:** 146
- **Toxicities:** Biomarker analysis; safety is reported in the parent cohort publications.
- **Efficacy:**biomarker 215 H-score; **biomarker** 98%

sacituzumab tirumotecan (sac-TMT) phase 3 randomized trial — NCT07419295 (TROP2 axis

Powles et al. *Ann Oncol* 2025

- **Disease context:**Advanced urothelial carcinoma after platinum-based chemotherapy and checkpoint inhibitor therapy (2L+) — Confirmatory trial for the accelerated approval; median follow-up 9.2 months.
- **n:** 711
- **Toxicities:**neutropenia (grade ≥ 3): 35.0%; febrile neutropenia (grade ≥ 3): 12.0%; **any treatment-related AE** (grade ≥ 3): 67.0%; **any treatment-emergent AE** (grade 5): 7.0%
- **Efficacy:**OS 10.3 months; **HR_OS** 0.86 (95% CI 0.73-1.02); **PFS** 4.2 months; **HR_PFS** 0.86 (95% CI 0.72-1.03); **ORR** 23.0% (95% CI 18.0-27.0); **TRAE_rate** 67.0%

Loriot et al. *Clin Cancer Res* 2024

- **Disease context:**Unresectable locally advanced or metastatic urothelial carcinoma (2L+) — Archival tumor from TROPY-U-01 cohorts 1-3; 146 of 192 treated patients had evaluable TROP2 data scored by H-score (0-300) and percentage membrane-positive tumor cells.
- **n:** 146
- **Toxicities:** Biomarker analysis; safety is reported in the parent cohort publications.

- **Efficacy:biomarker** 215 H-score; **biomarker** 98%

sacituzumab govitecan (TROP2 ADC) — trial-only after the 2024 urothelial withdrawal (NCT03547973 cohort 2 is the platinum-ineligible route)

Powles et al. *Ann Oncol* 2025

- **Disease context:Advanced urothelial carcinoma after platinum-based chemotherapy and checkpoint inhibitor therapy (2L+)** — Confirmatory trial for the accelerated approval; median follow-up 9.2 months.
- **n:** 711
- **Toxicities:neutropenia** (grade ≥ 3): 35.0%; **febrile neutropenia** (grade ≥ 3): 12.0%; **any treatment-related AE** (grade ≥ 3): 67.0%; **any treatment-emergent AE** (grade 5): 7.0%
- **Efficacy:OS** 10.3 months; **HR_OS** 0.86 (95% CI 0.73-1.02); **PFS** 4.2 months; **HR_PFS** 0.86 (95% CI 0.72-1.03); **ORR** 23.0% (95% CI 18.0-27.0); **TRAE_rate** 67.0%

Tagawa et al. *J Clin Oncol* 2021

- **Disease context:Locally advanced or metastatic urothelial carcinoma after platinum-based chemotherapy and a checkpoint inhibitor (2L+)** — Median three prior therapies; 66.4% visceral metastases. Entry required prior platinum, which this patient has not had.
- **n:** 113
- **Toxicities:neutropenia** (grade ≥ 3): 35.0% (n=113); **febrile neutropenia** (grade ≥ 3): 10.0% (n=113); **leukopenia** (grade ≥ 3): 18.0% (n=113); **anemia** (grade ≥ 3): 14.0% (n=113); **diarrhea** (grade ≥ 3): 10.0% (n=113); **treatment-related AE leading to discontinuation** (grade any): 6.0% (n=113)
- **Efficacy:ORR** 27.0% (95% CI 19.5-36.6); **DoR** 7.2 months (95% CI 4.7-8.6); **PFS** 5.4 months (95% CI 3.5-7.2); **OS** 10.9 months (95% CI 9.0-13.8); **discontinuation_rate** 6.0%

Tmb High interventions

1. nivolumab + ipilimumab checkpoint re-intensification — lower-ipilimumab schedule with protocolized amylase/lipase surveillance and written stopping rules; the nivo1/ipi3 boost is vetoed and stays off the table

Likelihood of effect: Uncertain: the 33% ORR (pmid:36868252) belongs to the vetoed ipi-3 boost in post-platinum, post-nivolumab patients; the tolerable schedule never beat monotherapy.

Toxicity burden: High (immune enterocolitis 11% grade 3-4 with 2/83 treatment-related deaths at ipi-3; pancreatitis rechallenge risk)

Counter-productive MoA:Moderate (Dose-reducing ipilimumab may delete the CTLA-4 contribution that ever beat PD-1 monotherapy.)

Overall:Prescribable now with no screening gate, but the efficacy evidence and both vetoes attach to the same high-dose schedule; the tolerable version is the unproven one.

Key references: PMID 36868252 · PMID 31100038 · NCT03866382

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

N-803 / nogapendekin alfa inbakicept (IL-15 superagonist) +/- NK cells — NCT03228667 cohorts / single-patient IND route — Thin evidence

Likelihood of effect: Unknown in mUC: no efficacy rows in the dossier; the rationale is mechanism plus the patient's prior deep IO response.

Toxicity burden: Moderate (cytokine-class constitutional and inflammatory events; pancreatitis-history watch)

Counter-productive MoA:Low (IL-15-driven expansion may re-stimulate an already-exhausted response only transiently.)

Overall:The immune-intensification adjunct the team is already exploring — no metastatic evidence base to rank it on.

Key references: NCT03228667

Evidence in detail

nivolumab + ipilimumab checkpoint re-intensification — lower-ipilimumab schedule with protocolized amylase/lipase surveillance and written stopping rules; the nivo1/ipi3 boost is vetoed and stays off the table

****Grimm et al. *Lancet Oncol* 2023****

- **Disease context:**Metastatic or unresectable urothelial cancer of the bladder, urethra, ureter or renal pelvis after platinum-based chemotherapy (2L+) — 83 patients, Karnofsky ≥ 70 , measurable disease; 50 (60%) received at least one ipilimumab boost, given either for early non-response or for later progression on nivolumab maintenance.
- **n:** 83
- **Toxicities:**immune-mediated enterocolitis (grade 3-4): 11.0% (n=83); **diarrhea** (grade 3-4): 6.0% (n=83); **treatment-related death** (grade 5): 2.0% (n=83)
- **Efficacy:**ORR 33.0% (95% CI 24.0-42.0); **CR_rate** 7.0%

****Sharma et al. *J Clin Oncol* 2019****

- **Disease context:**Platinum-pretreated locally advanced or metastatic urothelial carcinoma (2L+) — Three cohorts compared: nivolumab 3 mg/kg alone (n=78), nivolumab 3 plus ipilimumab 1 (n=104), nivolumab 1 plus ipilimumab 3 (n=92). Patients were checkpoint-naïve at entry.
- **n:** 274
- **Toxicities:**any treatment-related AE (grade 3-4): 39.1% (n=92); **any treatment-related AE** (grade 3-4): 30.8% (n=104); **any treatment-related AE** (grade 3-4): 26.9% (n=78); **pneumonitis** (grade 5): 1.1% (n=182)
- **Efficacy:**ORR 38.0%; ORR 26.9%; ORR 25.6%; **DoR** 22 months; **TRAE_rate** 39.1%

Personalized Neoantigens interventions

1. personalized neoantigen vaccine programs (peptide 2025 + mRNA 2026 series) — continuation of the already-running investigational adjunct, with the post-mRNA EliSpot as the branch point

Likelihood of effect: Low for objective response: no response evidence attributable to vaccination; the patient's own EliSpot sits at the weak end of the published distribution (pmid:37165196).

Toxicity burden: Low (injection-site and constitutional reactions in the trials on record)

Counter-productive MoA:Low (CD4-skewed responses without CD8 expansion may not translate to tumor killing.)

Overall:Already underway and preference-aligned; honest framing is hypothesis-stage, with the post-mRNA EliSpot as the branch point for any TIL/ACT escalation.

Key references: PMID 40346292 · PMID 37165196 · PMID 42223134

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

neoantigen-selected TIL + aldesleukin (NCI Surgery Branch) — NCT01174121 — _Not enrollable_

Likelihood of effect: Unassessable now: no resectable lesion for TIL generation after SBRT.

Toxicity burden: High (lymphodepletion plus aldesleukin class toxicity)

Counter-productive MoA:Low (None assessed; the surgical gate is the block.)

Overall:The ACT route the banked material points at, waiting on a lesion that no longer exists to resect.

Key references: NCT01174121 · PMID 40169866

Evidence in detail

personalized neoantigen vaccine programs (peptide 2025 + mRNA 2026 series) — continuation of the already-running investigational adjunct, with the post-mRNA EliSpot as the branch point

****Saxena et al. *Nat Cancer* 2025****

- **Disease context:**Urothelial cancer, adjuvant and metastatic settings (any) — 12 enrolled; vaccine successfully manufactured for 10 (median 20.3 weeks). Four treated in the adjuvant setting, five in the metastatic setting with measurable disease.
- **n:** 12
- **Toxicities:**injection-site reaction (grade 1-2): (n=10); fatigue (grade 1-2): (n=10); fever (grade 1-2): (n=10)
- **Efficacy:**biomarker 100%; ORR 40.0%; RFS; TTNT 20.3 weeks

****Rojas et al. *Nature* 2023****

- **Disease context:** Resected pancreatic ductal adenocarcinoma, adjuvant setting (adj) — 16 patients received atezolizumab and vaccine; 15 went on to mFOLFIRINOX. Vaccines were manufactured in real time from the surgical specimen.
- **n:** 16
- **Toxicities:** any AE attributable to the vaccine (grade ≥ 3): 6.0% (n=16); any AE attributable to the vaccine (grade 1-2): 100.0% (n=16); any AE attributable to atezolizumab (grade ≥ 3): 0.0% (n=19)
- **Efficacy:** biomarker 50.0%; RFS; RFS 13.4 months

****Khattak et al. J Clin Oncol 2026****

- **Disease context:** Resected stage IIIB-IV cutaneous melanoma (adj) — 157 randomized 2:1 to combination versus pembrolizumab alone; median planned follow-up 60.3 months at the December 2025 cutoff.
- **n:** 157
- **Toxicities:** No new safety signals at five years; per-term rates are not reported in this descriptive update.
- **Efficacy:** HR_RFS 0.51 (95% CI 0.294-0.887); HR_EFS 0.411 (95% CI 0.2-0.843); HR_OS 0.471 (95% CI 0.165-1.345); biomarker increased TCR clonality and novel clonotypes versus pembrolizumab alone

neoantigen-selected TIL + aldesleukin (NCI Surgery Branch) — NCT01174121

****Lowery et al. Nat Med 2025****

- **Disease context:** Treatment-refractory mismatch-repair-proficient metastatic gastrointestinal cancers (2L+) — Three sequential cohorts: 18 patients received bulk unselected TIL, 39 received neoantigen-selected TIL, 34 received neoantigen-selected TIL plus pembrolizumab.
- **n:** 91
- **Toxicities:** severe hematologic toxicity from lymphodepletion (grade ≥ 3): 100.0% (n=91); critical care support required (grade ≥ 3): 10.0%
- **Efficacy:** ORR 0.0%; ORR 7.7% (95% CI 2.7-20.3); ORR 23.5% (95% CI 12.4-40.0); biomarker response associated with more targeted neoantigens and more infused CD4+ neoantigen-reactive TILs

Prame Hla interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

brenetafusp (IMC-F106C, PRAME ImmTAC) — NCT04262466 / NCT07686367 — _Not enrollable_

Likelihood of effect: Unassessable: no enrollable cohort and the PRAME expression gate is unread.

Toxicity burden: Moderate (ImmTAC-class CRS and skin toxicity)

Counter-productive MoA: Low (None assessed; access is the block.)

Overall: The PRAME ImmTAC the field is actually developing — with no door open to a urothelial patient today.

Key references: NCT04262466 · NCT07686367

IMA203 / IMA203CD8 (PRAME-directed TCR-T) — NCT03686124 — _Thin evidence_

Likelihood of effect: Unknown until PRAME reads positive; published ORR 28.9% with mDOR 4.4 months is melanoma/sarcoma only (pmid:40205198).

Toxicity burden: High (lymphodepletion cytopenias, CRS ~universal in class, one septic death in the published series)

Counter-productive MoA: **Moderate** (Short response durability in the published cohorts; antigen-loss escape documented in class.)

Overall: A real PRAME door once expression reads positive — today it is double-gated with cross-tumor-only evidence.

Key references: PMID 40205198 · NCT03686124

IMC-P115C (half-life-extended PRAME ImmTAC) — NCT07156136 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; gated on the unread PRAME expression result.

Toxicity burden: Moderate (ImmTAC-class CRS and skin toxicity; no agent data)

Counter-productive MoA: **Low** (Same expression-gate uncertainty as the rest of the PRAME axis.)

Overall: The open-door PRAME option if expression ever confirms — nothing measured yet.

Key references: NCT07156136

Evidence in detail

IMA203 / IMA203CD8 (PRAME-directed TCR-T) — NCT03686124

****Wermke et al. *Nat Med* 2025****

- **Disease context:** PRAME-positive recurrent or refractory solid tumors in HLA-A*02-positive patients, including melanoma and sarcoma (2L+) — 27 patients in phase 1a escalation and 13 in phase 1b extension; 41 started lymphodepletion, 40 received IMA203. PRAME expression was measured by the sponsor's own RT-qPCR assay.
- **n:** 40
- **Toxicities:** CRS (grade ≥ 3): 4.9% (n=41); neurotoxicity (grade ≥ 3): 0.0% (n=41)
- **Efficacy:** ORR 28.9%; ORR 52.5%; DoR 4.4 months; biomarker deep responses enriched at higher PRAME expression

Mesothelin interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable

(program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CT-95 (mesothelin x CD3 bispecific) — NCT06756035 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data for the agent, a research-grade-only target call, and negative class precedents (pmid:35358455 class context).

Toxicity burden: Moderate (CD3-engager class CRS; no agent safety table)

Counter-productive MoA: **Moderate** (Soluble mesothelin sink and negative class precedent both argue the axis may not deliver.)

Overall: A mesothelin door that opens only after a validated stain — and the class record so far is negative.

Key references: NCT06756035 · PMID 40081946

TNhYP218 (anti-mesothelin TNaive/SCM CAR-T) — NCT06885697 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; the validated-stain gate and the class 0/20 precedent both stand.

Toxicity burden: High (lymphodepletion plus CAR-T class toxicity)

Counter-productive MoA: **Moderate** (Class CAR-T experience at this target has not produced responses.)

Overall: A next-generation construct aimed at a target whose urothelial case is still a research-grade stain.

Key references: NCT06885697

A2B694 (logic-gated Tmod mesothelin CAR-T) — NCT06051695 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; double-gated on a validated stain and the sponsor's screening genetics.

Toxicity burden: High (lymphodepletion plus CAR-T class toxicity)

Counter-productive MoA: **Moderate** (Same class negative precedent; logic-gate adds a genetics prerequisite.)

Overall: The most conditional mesothelin row: stain, screening protocol, and tumor genetics all unverified.

Key references: NCT06051695

SynKIR-110 (mesothelin KIR-CAR T cells) — NCT05568680 — _Not enrollable_

Likelihood of effect: Not applicable: histology-excluded.

Toxicity burden: High (lymphodepletion plus CAR-T class toxicity)

Counter-productive MoA: **Low** (None assessed; enrollment is closed to this histology.)

Overall: A closed door regardless of the stain.

Key references: NCT05568680

gavocabtagene autoleucel (gavo-cel, mesothelin TCR-fusion T cells) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: High (published class toxicity incl. severe CRS)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 37501016

huCART-meso (fully human anti-mesothelin CAR T cells) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: High (lymphodepletion plus CAR-T class toxicity)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 41566776

anetumab ravtansine (mesothelin ADC) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (maytansinoid class: corneal, hepatic, GI)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 35358455

M7A-DC / M7GA-DC (mesothelin DARPIn-MMAE conjugates, preclinical) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: High (MMAE class neuropathy, hypothetically)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 40849430

Evidence in detail

gavocabtagene autoleucel (gavo-cel, mesothelin TCR-fusion T cells) — unavailable

****Hassan et al. *Nat Med* 2023****

- **Disease context:** **Treatment-refractory mesothelin-expressing solid tumors (mesothelioma, ovarian, cholangiocarcinoma)** (2L+) — 32 patients dosed, 29 after lymphodepletion; 30 evaluable for response. Urothelial carcinoma was not among the enrolled histologies.
- **n:** 32
- **Toxicities:** **pneumonitis** (grade ≥ 3): 16.0% (n=32); **CRS** (grade ≥ 3): 25.0% (n=32); **bronchioalveolar hemorrhage** (grade 5): (n=32)
- **Efficacy:** **ORR** 20.0%; **DCR** 77.0%; **OS_rate** 70.0%

huCART-meso (fully human anti-mesothelin CAR T cells) — unavailable

****Barber-Rotenberg et al. *Mol Ther* 2026****

- **Disease context:** **Mesothelin-expressing lung adenocarcinoma, ovarian cancer and mesothelioma** (2L+) — 14 ovarian, 5 mesothelioma, 1 lung adenocarcinoma. No urothelial patients.
- **n:** 20
- **Toxicities:** **CRS** (grade any): 35.0% (n=20)
- **Efficacy:** **DCR** 60.0%; **ORR** 0.0%; **PFS** 12.3 weeks; **OS** 26.1 weeks

anetumab ravtansine (mesothelin ADC) — unavailable

****Kindler et al. *Lancet Oncol* 2022****

- **Disease context:** **Unresectable locally advanced or metastatic malignant pleural mesothelioma overexpressing mesothelin, after first-line platinum-pemetrexed** (2L+) — Prospective screening for mesothelin overexpression, defined as 2+ or 3+ membrane staining in $\geq 30\%$ of viable tumor cells; 248 of 589 screened patients were randomized.
- **n:** 248
- **Toxicities:** **neutropenia** (grade ≥ 3): 1.0% (n=163); **pneumonia** (grade ≥ 3): 4.0% (n=163); **dyspnoea** (grade ≥ 3): 6.0% (n=163); **any treatment-emergent death** (grade 5): 6.0% (n=163)
- **Efficacy:** **PFS** 4.3 months (95% CI 4.1-5.2); **HR_PFS** 1.22 (95% CI 0.85-1.74)

Muc1 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

DS-3939a (TA-MUC1-directed ADC, DXd-class payload) — NCT05875168 — _Thin evidence_

Likelihood of effect: Unknown: no clinical data; MUC1 call is research-grade only, and MUC1's ubiquitous normal-tissue expression makes the clone decisive.

Toxicity burden: Moderate (DXd-class payload: nausea, ILD risk — the active T-DXd liabilities)

Counter-productive MoA: **Low** (Ubiquitous low-level normal-tissue MUC1 could sink specificity.)

Overall:The only active MUC1 ADC — confirmation-first, and its payload re-runs the current drug's problems.

Key references: NCT05875168 · PMID 40635151

NEOK002 (EGFR x MUC1 bispecific ADC) — NCT07612189 — _Not enrollable_

Likelihood of effect: Not applicable while T-DXd is the most recent systemic therapy.

Toxicity burden: Moderate (bispecific ADC class profile; no safety table)

Counter-productive MoA:Low (None assessed; sequencing exclusion is the block.)

Overall:Closed by the current drug's payload class, not by the target.

Key references: NCT07612189

XYA02 (MUC1-directed agent) — NCT07670312 — _Not enrollable_

Likelihood of effect: Not applicable: study not open, ex-US.

Toxicity burden: Moderate (unknown; no data)

Counter-productive MoA:Low (None assessed.)

Overall:A door that has not opened, on another continent.

Key references: NCT07670312

gatipotuzumab (anti-TA-MUC1 antibody) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Low (naked-antibody profile)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 35397434

Ad-sig-hMUC1/ecdCD40L adenoviral vaccine — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Low (vaccine-class reactions)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 36307410

MUC1-C inhibition (GO-203 class) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (unknown; no current clinical program)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Evidence in detail

gatipotuzumab (anti-TA-MUC1 antibody) — unavailable

Ochsenreither et al. *ESMO Open* 2022

- **Disease context:**Refractory EGFR-positive metastatic solid tumors, then an expansion in colorectal, non-small-cell lung, head and neck and breast cancer (2L+) — 20 patients in the primary phase and 32 in the expansion phase. Soluble TA-MUC1 was tracked in serum as a candidate companion biomarker.
- **n:** 52
- **Toxicities:**dose-limiting toxicity (grade any): 0.0% (n=52); **related serious adverse event** (grade any): 0.0% (n=52); **related death** (grade 5): 0.0% (n=52)
- **Efficacy:**ORR 9.6%; **biomarker** above-median baseline soluble TA-MUC1 associated with longer PFS and OS

Ad-sig-hMUC1/ecdCD40L adenoviral vaccine — unavailable

Tan et al. *Nat Commun* 2022

- **Disease context:**Advanced adenocarcinoma (breast, lung, ovary) (2L+) — 21 patients with advanced adenocarcinomas; the trial was not designed to determine clinical efficacy.
- **n:** 21
- **Toxicities:**injection-site reaction (grade any): 71.0% (n=21); **dose-limiting toxicity** (grade any): 0.0% (n=21)
- **Efficacy:**AE_rate 71.0%; AE_rate 0.0%

Fap Stroma interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

FAP-targeted radioligand therapy (FAP-2286 / FAPI class) — NCT04939610, gated by FAP PET — _Thin evidence_

Likelihood of effect: Unknown: case-series evidence only (pmid:34385340), and the one FAPI PET on record showed no FAP-avid tumor.

Toxicity burden: Moderate (radioligand class marrow and off-target stromal exposure)

Counter-productive MoA:Low (Stromal targeting without tumor avidity would treat inflammation, not cancer.)

Overall:An imaging question wearing a therapy costume: no FAP-avid urothelial disease has ever been shown in this patient.

Key references: PMID 34385340 · NCT04939610

Evidence in detail

FAP-targeted radioligand therapy (FAP-2286 / FAPI class) — NCT04939610, gated by FAP PET

Ferdinandus et al. *J Nucl Med* 2022

- **Disease context:**Metastatic soft-tissue or bone sarcoma (n=6) and pancreatic cancer (n=3) after exhaustion of approved therapy (2L+) — Selection required high FAP expression, defined as 68Ga-FAPI-46 PET SUVmax ≥ 10 in more than 50% of all lesions.
- n: 9
- **Toxicities:**thrombocytopenia (grade ≥ 3): 44.0% (n=9); anemia (grade ≥ 3): (n=9)
- **Efficacy:**DCR 50.0%; biomarker 1.28 Gy/GBq; biomarker 78.0%

Arid1A interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

ART0380 (ATR inhibitor) — NCT04657068 (ARID1A rationale) — _Thin evidence_

Likelihood of effect: Unknown: hypothesis-grade rationale; class clinical responses have clustered outside this histology.

Toxicity burden: Moderate (ATR-class cytopenias)

Counter-productive MoA:Low (Monotherapy ATR inhibition in a tumor without measured replication stress may do nothing.)

Overall:A hypothesis-grade door that does not require its own hypothesis to enter.

Key references: NCT04657068 · PMID 27958275

tuvusertib (M1774, ATR inhibitor) + avelumab — NCT06518564 (ARID1A-selected, endometrial-restricted) — _Not enrollable_

Likelihood of effect: Not applicable: endometrial-restricted; informational for the ARID1A axis only.

Toxicity burden: Moderate (ATR-class cytopenias plus checkpoint combination irAE risk)

Counter-productive MoA:Low (None assessed; enrollment is closed to this histology.)

Overall:The ARID1A trial that actually selects for ARID1A — closed to this histology.

Key references: NCT06518564

PARP inhibition (+/- temozolomide) for ARID1A-mutant disease — off-label, hypothesis-grade — _Thin evidence_

Likelihood of effect: Unknown: hypothesis-grade preclinical rationale with no supporting biomarker in this record.

Toxicity burden: Moderate (PARP-class cytopenias and fatigue)

Counter-productive MoA:Low (Without HR deficiency the mechanism has no lesion to exploit.)

Overall:A borrowed synthetic-lethality argument waiting on a somatic HR readout that already exists in an unretrieved report.

Key references: PMID 37306706

EZH2 inhibition (tazemetostat) for ARID1A-mutant disease — off-label, hypothesis-grade — _Thin evidence_

Likelihood of effect: Unknown: preclinical-only rationale in this context.

Toxicity burden: Low (tazemetostat class profile is mild: fatigue, cytopenias under 20% grade 3)

Counter-productive MoA:Low (Epigenetic reprogramming without a measured dependency may select resistant states.)

Overall:The mildest drug on the weakest rationale in the case.

Key references: PMID 32506298

ceralasertib (AZD6738, ATR inhibitor) — ARID1A rationale — _Not enrollable_

Likelihood of effect: Low: responses confined to gynecologic histologies (pmid:41686845); nothing supports carryover.

Toxicity burden: Moderate (ATR-class cytopenias)

Counter-productive MoA:Low (Same unmeasured-dependency problem as the rest of the ARID1A axis.)

Overall:Cross-tumor evidence for someone else's tumor type.

Key references: PMID 41686845

berzosertib (M6620, ATR inhibitor) — unavailable — _Unavailable_

Likelihood of effect: Not applicable: no accessible program.

Toxicity burden: Moderate (ATR-class cytopenias)

Counter-productive MoA:Low (None assessed; the program is not available.)

Overall:Not reachable by any enrollment or access route on record.

Key references: PMID 32568634

Evidence in detail

ceralasertib (AZD6738, ATR inhibitor) — ARID1A rationale

****Zhu et al. *Clin Cancer Res* 2026****

- **Disease context:**Locally advanced or metastatic solid tumors with ARID1A loss by immunohistochemistry (2L+) — 29 efficacy-evaluable patients, all with radiographic progression at entry and ARID1A loss confirmed by IHC rather than by sequencing alone. 13 had gynecologic malignancies.
- **n:** 29
- **Toxicities:** Safety detail is in the full text rather than the abstract; ceralasertib monotherapy toxicity is dominated by reversible myelosuppression at this schedule.
- **Efficacy:**ORR 14.0%; ORR 33.0%; DCR 31.0%; DoR 33.7 months

berzosertib (M6620, ATR inhibitor) — unavailable

****Yap et al. *J Clin Oncol* 2020****

- **Disease context:**Advanced solid tumors, unselected for ARID1A (2L+) — 17 patients received monotherapy and 23 the carboplatin combination. Molecular annotation was retrospective rather than an entry criterion.
- **n:** 40
- **Toxicities:**hematologic toxicity (combination with carboplatin) (grade ≥ 3): (n=23)
- **Efficacy:**CR_rate 2.5%; DCR 42.5%